

ActInf Livestream #004.1 "Cultural Affordances: Scaffolding Local Worlds"

Livestream | 1:37:46

live streaming

hello everyone and welcome this is team

com podcast 4.1 on september 15th 2020.

welcome to everybody hello all

participants

i'm going to

share my screen and we will jump right

in

this is team com podcast 4.1

and this is team com

teamcom is an experiment in online team

communication and learning related to

active inference

you can find us on twitter at

inferenceactive

you can email us you can find us on the

public key base team

as well as at the youtube channel

this is a recorded and archived live

stream

so please provide us with any feedback

participants as well as viewers so that

we can improve our work

all backgrounds and perspectives are

welcome here and also just on a note of

video etiquette if people are muted in a

noisy background or raising their hands

when there's this many participants in

the chat it would be excellent and help

us with our turn taking and reduction of

uncertainty about who's to speak next

so here we are in stream 4.1 and the

structure of today is going to go like

this first we're going to have a few

check-in questions and warm-up

discussion

then we'll get to the paper itself which
is ramstead at all 2016.

we'll talk about the big aims of the
paper

go through a roadmap of the paper to see

how the authors get from a to z

we'll go through the abstract to

understand how the authors represent
their own work

then we'll go through the box of

definitions as well as the figures

at that point we'll have time for

questions and other insights and also

we've implicitly prepared for a podcast

4.2 where we'll be able to have further
discussions on some key quotations or

anything arising that we don't get to

this week

so

thanks for that part here we are in the
check-in and everybody can feel free to
introduce themselves in their location

if they would like the first question is

pretty broad but that's what the warm-up

is about so what is culture

it's a good question

or in the sense of this paper in the

sense of culture that's being scaffolded

within a kind of framework that we might

want to build out what would that kind

of culture refer to

well so i'm maxwell ramstad i'm uh the

first author on this paper cultural

affordances

um by the way i'm really grateful uh for

this uh podcast you all have been so

great just in general over the last few

weeks and i'm i'm particularly touched
that you're giving uh the work that
we've developed so much attention um and
it's like just i'm just so full of
gratitude

this is so cool um so culture
at least in the sense that we're using
uh
the term here means something like the
taken for granted

i think is sort of the heuristic way
that i like to think about it it's sort
of like in a given local frame of
reference

uh in a given environment shared by a
group of people a living together what
are the things that are taken for
granted what are the
uh

sets of expectations and beliefs and
sets of practices and norms
um

you know that that are shared and that
essentially uh structure people's
intuitive grip on their shared social
world

any other thoughts on culture there yeah
in sociology uh we define culture it's
kind of three different dimensions
you have the symbolic ideal dimension
which is really i think equivalent to or
equivalent to semantic knowledge uh we
break it up in terms of norms values and
beliefs

uh we have a psychobehavioral dimension
which is really about in body culture
how your body moves um the skills that
you have in in terms of psychological

kind of terminology i would really say
your skills and habits um in in kind of
that non-declarative memory sense and
then the third dimension is really our
material dimensions of the physical
things that we create the tools the
buildings the fashion

um which i think

the degree to which i understand the
affordance literature i think all three
of those dimensions definitely have
versions of affordances that could
connect to that um

i think that was one thing i did see in
the paper that it did seem to be very
much focused on just that semantic
kind of symbolic dimension and not
really uh explicitly incorporating the
other dimensions that at least in
sociology we would kind of um include in
that definition of culture

cool

yeah thanks for that that was very rich
actually uh

very interesting uh points that you're
raising yeah i should just give some
maybe some historical background on this
paper this was the first in a series of
papers that have been developed into
basically like a a variational or free
energy approach to culture

and i think a lot of the elements that
you're mentioning were only sketched in
this paper uh and were developed in i
think further iterations of the model
uh so everything that you're that you
just outlined uh richard i think are are
very essential and crucial things that

need to be

tackled when studying culture so it's

great that you're raising them

cool and i think the question about

multiple kinds of culture ranging from

the semantic concepts to the

externalized productions or the

artifacts of culture

bring us this idea of scaffolding

culture so what is a way that culture is

scaffolded

someone can raise their hand maybe

someone new but something that guides

the construction just like a scaffold

i'll give one way which is education

because pedagogy is sort of the guide

rails for how individuals learn about

those norms as well as how to use the

artifacts and construct new artifacts

these are ways in which culture is

scaffolded because it provides structure

for the growth to happen within

any other comments on that

okay and then the last warm-up question

is what is shared intentionality or

what's a way in which somebody has seen

shared intentionality play out in their

recent experience

well so there are two parts to that

right there's shared and there's

intentionality so intentionality at

least in the sense that we're using it

in the paper draws on the philosophical

construct of intentionality that we we

can

basically uh

you know

you can draw a lineage all the way back

to brentano effectively but so by
intentionality we mean orientation
towards the world
uh we don't necessarily mean
intentionality in the sense of uh having
the intention to do this or that
it really
it's from the latin i think intensio
which is the word that was used to
describe what a uh
what an archer is doing with his arrow
uh to hit the target it is intensio with
the target and so it
it it's this that we're trying to get at
and one of the things that we're trying
to argue here is that what part of what
makes um a shared social world just that
a shared social world is that
there are forms of relating to that
world so forms of intentionality that
are also shared
uh among the the people who live in that
world together so that's that's what we
were trying to get at ultimately i think
and i think what's interesting in terms
of if you take that definition and you
look back at the scaffolding um you know
if you look within the individual
there's this network of semantic and
episodic memory and non-declarative
memory that's organized in the brain
if you put me in contact with another
individual from my culture who shares my
language and we begin to construct this
we space if you look at the
phenomenology um and then now what
happens with this dyad
either we introduce a third person or we

then one of us leaves the relationship
and we go uh interact with someone else
if you spread that dyadic interaction
out to a network and then you look at
the fractal scaling of that network
there's this way to i think to very
deliberately keep track and build up
from organized memory and brain
to an institution of 25 000 people
dynamically interacting by following
this

this these mechanisms by which you
create the shared space and looking at
those emerging properties and trying to
model them mathematically

couldn't agree

and there's a quote from the paper which
i'll spare maxwell from reading is uh
near the end it says to address more
complex social situations it is useful
to revise current sociocognitive models
of joint attention to encompass
fundamentally triadic situations in
which the third

is the socially constituted niche of
affordances supported by local
ontologies and abilities so we're just
jumping in today but that's sort of
where we're wanting to go with this
paper and in discussion is towards
understanding what is that third space
and how does the aim or does the intent
play in yeah stephen

yeah and i think this cultural thing
also brings in that into subjectivity
it gives a way to
understand how to approach it beyond
education

beyond
sort of the banking of
programming people saying well what's in
the environment particularly now that
we've got situations of
cross-cultural
what cultures have to try and
understand and work together is it all
about teaching us how to tolerate each
other
or is it about understanding
how we perform in the environment
so i think it's got a lot of
well here we are finally at the paper
uh cultural affordances scaffolding
local worlds through shared
intentionality and regimes of attention
so we'll definitely want to unpack each
of these concepts and it's in frontiers
in psychology from 2016.
the big aim of the paper grasped from
the abstract in the
introduction section is to better
understand how culture and context
interact with human biology to shape
human behavior cognition and experience
so kind of all of it kind of things that
richard was bringing up early on
and from the introduction the
integrative framework we propose bridges
cognitive and social sciences to provide
two things first an expanded concept of
affordance that extends to
socio-cultural forms of life which will
unpack in box one and two a multi-level
account of the socio-culturally
scaffolded forms of affordance learning
and the transmission of affordances in

pattern sociocultural practices
so the key piece there is that
multi-level account because we're not
going to have an individual level
account by which collective outcomes
arise or a group level account by which
individuals are shaped in their behavior
uniquely those are sort of two sides of
the same coin or
some of the sides of a many-sided coin
that constitute this top down and
bottom-up relationship that we see in a
lot of multi-scale systems like the
brain being nested within a body inside
of a niche inside of these institutions
that last beyond the lifespan of the
organism
any thoughts on the aim here
or questions about anything
cool well then let's get to the roadmap
so the roadmap i just copied out the
section headers
how that takes us on a journey or on a
narrative in our learning odyssey in a
sense
in the beginning there's an introduction
and a theoretical framework for
affordances is provided and that
includes i know the font may be a little
small perspectives importances and
phenomenology
landscapes and fields meanings and
affordances and then we reach figure one
which has a basic cognitive formula for
automatic intentionality and figure two
that expands on that formula slash
comment format so that's a theoretical
framework for affordances maybe could

anyone chime in what what is an affordance in this uh context or in the cultural context or in the context of whichever field you're familiar with yeah so i guess just for a simple definition it's something that invites action or interaction and the the cartoon example is always a chair a ford sitting um but in the context of a social interaction it could be some environment that affords or that elicits our interaction maybe conversation or music making with another person great and it makes sense to start the paper off with this framework for affordances because action opportunities are so fundamental to active inference and to action-oriented frameworks this is really what defines action-oriented frameworks is they focus on actions and the stimuli or the cues that lead to action so affordance is their opportunity the second section here is the neurodynamics of affordances so this is the integration of the cognitive and the neurosciences with the more behavioral ecological concept of an affordance computation representation and minimal neural models is addressed and also many of these ideas later papers include other or different depictions and this is already four years past so that equates to about 400 papers

in between us and here

yeah i should say um like this is

probably the weakest section of the
paper as it currently

is formulated

um

i mean yeah as you're pointing out it

was published in 2016 and we started

really working on this in 2015

and i mean since i did a whole phd

focused on the free energy principle and

uh what it what it entails and how it

works and so on and i think uh

i i mean i wouldn't disavow what we say

here uh

but i think that

definitely our our understanding of the

representational implications

of the theory have been refined over the

years but i mean this this is basically

in a nutshell what we tried to pursue

since 2016 so as you're saying like

several papers have been published since

um and the uh you know

the the the framework has moved a little

bit

absolutely richard

yeah i think i mean definitely whenever

we look back on our earlier papers we're

always like i should have said all these

different things but i i actually really

appreciate this paper because you know

for reading it just as a complete

non-expert in

a lot of these fields um the free energy

principle is very very difficult to find

an entry point into

and sort of have something that i think

is very conceptual and grounded in neuroscience i think a lot of social sciences could wrap their head around you know relatively easily and then to trace this as breadcrumbs to the more advanced

um free energy principle stuff um i i really appreciate i thought it was a great entry point into this literature and kind of gave me a you know a foundation to kind of start to wrap my head around how these these concepts are being applied in that more mathematical context

you're saying that

it's one of the reasons why we chose this paper other than shannon suggesting it is for me at least when i can see the path in the literature well okay here's what they thought in 2016 which means in years leading up to 2016

and then yes it can be refined or represented or enriched in later work but seeing the breadcrumbs or the pheromone trail is how i get a calibration of the derivatives of how the ideas are changing through time and i found that um yeah one way maybe weaker but the most growth has happened here and this is one of the areas where the most growth can happen so this is why we continue to dip back into the literature

even longer than four years in the past i know it feels like forever though because sometimes there's a uniqueness to that cultural artifact that isn't just about how well it conveys what the

relationship is between neuroscience and free energy but rather how well it conveys the what the researchers were working on in that intersection so we're thinking through other minds for some of us it was us in the past for some of us it was just another paper yeah and i will say this just really quick too

a lot of the affordances that concept of affordance in sociology is very much grounded in kind of a pragmatic process philosophy you know whitehead the idea that you know reality reaches out to the other reality and so if you if that is where our literature is grounded to be able to have something like this that's very conceptual to work through those philosophical ramifications and then begin to ask how would this fit with uh the free energy principle and i think vice versa as well having the free energy principle community kind of understanding some of these philosophical concepts and ideas to kind of ground more conceptually not so much mathematically what it is that their perspective is offering to other disciplines

[Music]

well said

then we move to figure three now i found that figure three four and five make a really nice sequence and we'll go through them in sequence because it enriches one level of detail on top of one level of detail of a hierarchical prediction error minimization framework

figure four diagrams it a little bit
more clearly from a bayesian perspective
and then five brings on another level of
realism
then we talk a little bit in this
section about predictive processing and
attention and that returns our attention
to the idea of culture as regimes of
attention from the title
at this point if i had to rewrite this
paper this would be two papers and this
would be the split
so i really think that the first part of
the paper really uh tells a story about
affordances uh affordances the
affordances that are avail available in
the human landscape uh you know in terms
of their depend the dependence of
certain kinds of affordance on
convention whereas others are more just
dependent on the kind of spontaneous you
know action biology of human bodies and
then the second half of the paper turns
to this concept of attention
and with with all this in place says
okay so how is culture acquired in these
what we're calling regimes of attention
which are effectively shared patterns of
allocating your attention to this rather
than that as being salient so and this
all is bit you can read this as kind of
computational goffman uh for richard
who's done comes from sociology uh yet
it's an attempt to kind of spell out
how
what would a mechanistic model of you
know regimes of like civic inattention
and so on look like effectively uh when

you're trying to bridge these different levels

but you know i think there's a lot in this paper and this is a good kind of split point if you want to see what the paper is doing effectively

nice very helpful for us because it just reads as a turing tape to those who are reading it line by line but to know that there's a conceptual demarcation is helpful

and regimes of attention i just think is a great term to capture it because it's kind of like well where's the regime it's everywhere and it's nowhere it just is the water that culture is existing within

in this section we have figure 6 of a diagram of the looping effects or the rolling loops

that mediate cultural affordance learning

there's a discussion of skilled intentionality and affordance competition with some nice citations to other work about the skilled intentionality framework

there's a section on shared expectations local ontologies and cultural affordances and we think a lot about ontologies here and about the words that we use to locally organize our thought and cultures this kind of multi-scale ontology that helps organize stimuli in a sense

and the section closes with a discussion of shared expectation and implicit learning

then the last
section major section of the paper is
the regimes of shared attention and
shared intentionality
reflecting gating abilities and
affordances
and moving the loop regimes of shared
attention and skilled intentionality
before a
conclusion so that is the roadmap that's
how you get from discussion of the
ecological psychological or behavioral
inactive concept of an affordance
then into the neuroscience of
affordances and then as suggested by
maxwell there's a bit of a intermission
conceptually here before turning to a
second half of the paper that has to do
with the actual ways in which culture is
scaffolded through these regimes of
expectations and attention
any questions on the roadmap
cool

all right here we are in the abstract
which i often find helpful to read
because i know that authors put a lot of
care into the abstract because it
represents to the in group to the field
as well as to the out group to the
reviewers to the editor to the reader it
reflects what the paper is going to be
about and people get surprised and even
disappointed when the abstract is off
base relative to what the paper actually
delivers on
they write
in this paper we outline a framework for
the study of mechanisms involved in the

engagement of human agents with cultural affordances which we're going to define soon

our aim is to better understand how culture and context interact with human biology to shape human behavior cognition and experience

we attempt to integrate several related approaches in the study of the embodied cognitive and affective substrates of sociality and culture and the socio-cultural scaffolding of experience

that's the first half the abstract

then they write

the integrative framework we propose is

we propose bridges cognitive and social sciences to provide

one an expanded concept of affordance that extends to sociocultural forms of life and two a multi-level account of the socioculturally scaffolded forms of affordance learning and the transmission of affordances in a patterned socio-cultural practices and regimes of shared attention

the framework provides an account of how cultural content and normative practices are built on a foundation of contentless basic mental processes definitely

something we'll return to

the acquire content through immersive participation of the agent in social practices that regulate joint attention and shared intentionality

okay

any

thoughts on the abstract or questions

before we turn to the box

cool

well box one i thought it would be helpful just to list out the terms of the different types of affordances then go through them what is an example of each of these kinds of affordances or what are the differences between these types of affordances so maybe we could start there if someone could raise their hand and pick one of these types of affordances natural cultural or conventional and then define uh what it means to them or an example of it yep maxwell

um well essentially what we're suggesting is that well cultural affordances by the way um i i just noticed that my um my camera is frozen is that the case for everyone as well your screen looks dark you can either toggle the camera on and off or just reload the page um

after talking i might try to rejoin just so the camera freezes um yeah so what we're essentially arguing is that in humans all affordances are at least minimally cultural

in the sense that i mean i don't know about you but as i was growing up uh my mother told me not to slouch uh while i was at at the dinner table

uh and so you know it's something as simple and natural as

sitting um does actually come with a cultural scaffold and what we're saying is that there's basically a spectrum of affordances

you can think about the space of

affordances with his with which humans interact as going on a kind of continuum from natural to conventional where natural affordances depend mainly just on the biophysics of the human body in its interaction with the environment whereas conventional affordances depend on these implicit or explicit sets of uh conventions or shared beliefs uh so you know an example of the latter would be um a king's throne right you don't sit on the king's throne unless you're a very specific agent uh so a king's throne doesn't afford sitting

uh for everyone in the same way that say just a chair or a stool might afford sitting

there there are conventions shared conventions

uh the kind of uh structure which affordances are available uh and which are not for a given agent cool richard

yeah i was just curious in the neuroscience about like actually distinguishing that so if i if i see a cup my mirror neuron system fires and i begin to kind of enact the act of reaching out and grabbing it but then semantically i know that you know as a a lowly surf i'm not allowed to touch that is there actually any research showing the conflict between semantic knowledge about you know and self-related knowledge that i'm not allowed to do that with the automatic kind of reaching out

mirror neuron system simulation of
touching the cup
that's really interesting i'm not sure i
i don't know if there's any
data about this particularly it would be
very interesting though
one thought i had was some experiments i
believe by michael von elk with hands
grasping cups at incorrect ways so it's
something when you're here and you're on
c oh that's not how you're supposed to
grab a cup well why not well because i
can't rotate my wrist again to drink it
so it's the wrong way to hold it because
it's not really a cup of water if i
can't drink from it in that functional
affordance based sense but i agree it
would be interesting and we'll follow up
on that and stephen did you have any
class
yeah i actually had a workshop when i i
was working with a group with um who
with disability who um
and i mentioned about affordances as
part of it and partly having done this
paper and one of the participants she
was saying how when she sees a cup
because she can't pick it up and she
doesn't see it as a cup unless it's one
that's got a handle that she can reach
her hand into
and do that so that kind of
sense of things presenting is quite true
and i like this breakdown because
the i get a sense that the following on
from the later work that maxwell did
like natural affordances is like you
know if there's a cliff it doesn't

afford me to walk any further whereas
cultural is a bit more of like this
mixing sort of egotic
kind of space where it's conventional
it's like the the throne it's like it's
not up for negotiation it's like you
don't sit on the chair it's like a fixed
rule there's no
kind of
inter-subjective piece going on so in
some ways the cultural piece seems to
have some sort of negotiation associated
with it which is quite quite cool
would it be accurate to say that that in
a sense of if i
i had the simulation to reach out and
grab the cup but then my semantic memory
says no we're not allowed to do that
could that almost be like a prediction
error in the sense of like i'm
predicting that i'm going to do this
well no we don't do that precisely and
the way that you would do that is
probably by having a hierarchical model
where basically you know so the the
notion of hierarchy i mean it has
political implications and we can kind
of ditch those
uh so
i prefer thinking about it in terms of
center and periphery right where like
the the periphery is the the end of it
that's closer to the sensory motor like
interactions and the center is something
more like you know what we would think
of as the top of uh of a processing
hierarchy where representations are
progressively more compressed uh so

if you look at the hierarchy from from
periphery to center or from bottom to
top what you have represented there uh
if you'll allow me the

language of representation for a second
we can finesse it later but what's
essentially represented there is as you
progress towards the center slower and
slower or more and more abstract
regularities

so you might think that yeah as you see
the chair at a kind of more automatic uh
kind of a sensory motor level you you
see it as something that you could sit
on but as that ascends the processing
hierarchy at some point it registers an
error as in like semantically i'm not
allowed to sit on this because it's a
king's chair

so precisely that's how that would work
as and you can you can accommodate for
this kind of conflict by allowing the
system to have a layered or hierarchical
structure

and

great

point and this relates to how it's a
spectrum of affordances so for example a
violin when i look at a violin i have
the biophysics to bend my fingers but i
can't play violin so i see an affordance
for something i could pick up
potentially and not everybody may see
that affordance however somebody else
sees that and culturally they think oh i
could play this

uh song or i could play that song and so
what song they see the device as being

able to be used for is going to be part
of their enculturation part of their
education which builds upon the
biophysics of the human body and that's
why we have different dance styles in
different places
from different cultures even though
everybody's elbows still move the same
way sasha
yeah
so in trying to understand affordances
um i
tend to think of it as like the space of
what is allowed
but in walking through the different
kinds of um affordances that we have um
it becomes pretty clear that it's not
just what is allowed it's
very strongly gated by the top-down
priors of what you think should be
allowed in your space
and it just kind of reminds me of this
um
anecdotal understanding of how children
interact with an environment versus
adults
and
how children would use or sit on objects
that adults would never think of sitting
on
and kind of the the different ways that
they
see and use the world um because they
don't have these top-down priors of um
what one should and should not do
and so that that's a nice way of um kind
of circling back to the point of this
paper that um everything is embedded in

culture and you don't just start from
kind of the biophysics of sitting and
work your way up
you're always going back and forth with
this is uh this is such a great point
thank you for thank you for uh pointing
that out sasha like the
the agenda here is to
uh pursue this kind of
excuse me this kind of cooperative turn
in the anthropology literature which is
to say that
um
you know as joseph henrich who's one of
my uh one of my favorite like
anthropologists uh psychologists uh out
there as he points as he puts it we've
crossed the evolutionary rubicon in a
way our our survival as humans depends
on our ability to reliably access and
use
uh stored cultural information
effectively
and that's had an influence on the way
that our biology has evolved so you know
compared to chimps for example we have
uh really weak digestive juices
and that's because we've we've been able
to offload the metabolic cost of you
know having this extremely elaborate
digestive system
uh to our technology effectively you
know so tool use the use of fire cooking
technology has allowed us to process
food in a way that then co-evolves with
our bodies and takes the kind of
processing load of the food uh off the
body itself towards these cultural

artifacts and practices that have
co-evolved with our biology and that's
really the the point is that uh all
human affordances are cultural
affordances which exist on a spectrum
from more natural to more conventional
but human biology is a cultural biology
and biologic and
yeah culture is a biological phenomenon
i think that the two are kind of
irreconcilably joined in the case of
humans
and i'll
build on that to describe the difference
between the field and the landscape of
affordance sorry
i meant insurmountably you're uh
irreducibly joy not not here uh
yeah sorry
oh yeah it works so the field of
affordances are the affordances in the
landscape with which the organism can
dynamically cope and intelligently adapt
so that's at an organismal level the
field of affordances is like the field
of play it's the field of action for
that agent
the landscape of affordances on the
other hand like the landscape exists
beyond me even when i can't interact
with it directly and similarly the
landscape of affordances
is defined here as the total ensemble of
available affordances for a population
in a given environment and so the
concept of chair just as sasha was
bringing up the landscape of affordances
of chairs includes things for little

people for tall people for people with
small hands for larger hands the
landscape of affordances is at the
population level which means it includes
all these differences that are
biophysical and developmental and
cultural as well so the field is what
one individual agent is able to interact
with and that's sort of where the rubber
hits the road of action-oriented
frameworks and then when we pull back a
level to this evolutionary perspective
on action-oriented frameworks we get to
this landscape of affordance's idea
any thoughts on that

yes stephen

yeah i think that

that

way that things are embedded it comes
back to that piece with the throne as
well is

where

where that chair is for it to be
ascribed as a throne that you don't sit
on

is partly processed by what's going on
around

in that setting at that moment like if
there's no one around and it's an empty
room

you may sit on the chair if you don't
think anyone's going to see you but when
the cultural

kind of rules of the game are in play

or if you

see you know if it was a room where
there's only children in there compared
to a room where there was an adult

watching so

there's this kind of negotiation going
on

um which is quite

quite complex

and i think previously without free
energy i don't think there was a way to
be able to take that many different
parameters

and sort of integrate them

richard

yeah i think what's really fascinating
too when we talk about this is that um
the role of language in the role of
imitation and the role of just watching
other people where i can walk into a
room and know absolutely nothing about
the norms values and beliefs that that
constrain it but if somebody just tells
me you know four or five words they can
immediately bring me up to speed and
kind of and to use the terminology that
sasha was using earlier those top-down
priors i can gain a lifetime of
experience in 2500 words and watching
four or five people

seeing who's rewarded who's punished and
i think that i i i'm curious to see and
we don't have to talk about this now but
just you know rhetorically i'm curious
to see

how can you take in that notion of
symbolic communication and quickly
affect somebody's you know prior
understanding of situation how do you
fit that into a network of free energy
or agents running off this free energy
principle and how does that is it like a

synchronization of oscillators on a
network like how exactly do you model
model that process

yeah i'm just imagining you're walking
with somebody to a meeting and right
before they open the door to reduce your
uncertainty about what's behind the door
they say hey it's going to be tense in
there or hey just so you know it's a
non-smoking room and just like you're
saying with four or five words the way
that that person unpacks
those words symbolically and the way
that that communication channel is
utilized to attune the agents on a
narrative level

that's what dictates how well the scene
is going to go when the room is actually
being inactive and so there's so much
that we tell each other when we're
leading into events that that does set
the tune and that has to be unpacked
within each person's own perspective
and also one bonus word that's not here
but

umwelt does anyone have a sense of what
that means but i figured i just throw it
out there as they sort of complement to
affordance

well if i can just uh comment on uh
richard's point uh before getting into
unveiled i think the way that you can
understand it from a free energy
perspective is uh i mean andy clark i
think and andrus robstorff has have
called this top top

effects

so

basically uh passing information from the top or the center of one's hierarchy to the top of another directly through these kinds of uh of linguistic tokens that we're exchanging i think that's uh that's a powerful intuition what we've done in more recent work is extend that to a kind of horizontal top kind of communication where it's literally deconstruct the shared physical structure of the niche whether it's in the in terms of shared artifacts like stories that we uh create and and share with one another that so these uh effectively these uh parts of models that are offloaded onto the physical structure of the environment can also come and have these uh kind of uh top-down structuring effects where we're we're effectively installing shared environmental priors at into the top levels of our hierarchy through enculturation

nice and how about maxwell what's your take on him belt

i mean i think you know so i think those of us uh you know working in uh more specifically on ecological psychology we'll have some beef with the way that i kind of uh just generally put all of this together but i think like you know affordances umvelts uh phenomenology

i'm sure many people are not going to agree with me so you know feel free to discuss this but that they're they're all perspectives on the same kind of

thing which is that if we're trying to
talk about you know the meaning making
of organisms living organisms that share
a world we can't uh we can't just not
talk about the first person perspective
and the way that meaning appears from a
specific vantage point
and that's sort of what the affordances
and umwelt stuff gets at
uh and effectively what we tried to do
in in this paper
and that's why i think it's it's the
kind of contribution to the broader
project of neurophenomenology is we we
start from
the kind of first-person
perspective so we we're kind of trying
to cash out you know some some subtle
social effects like enculturation in
terms of learning affordances and then
we ask the question what's going on
under the hood and as it were around the
hood uh
for that kind of affordance story to be
cashed out in mechanistic terms
uh so umwelt is a term uh by a biologist
i think it's pronounced von ushakul
yeah and he uh
i mean it's it's essentially it's very
close to the uh
the uh
affordances story although it's it's
more directly centered on um
the perspective of the organism i mean
so gibsonians i should say like uh you
know the ecological psychology was
introduced by uh you know jj gibson the
famous psychologist and a lot of

ecological psychologists have uh take
issue with our use of affordances here
uh i think for interesting reasons the
so the for for ecological psychologists
the idea is that perception is the
direct perception of a possibility for
for well
it's it's the directly we're directly
picking up information about our
environment like so they're trying to
emphasize that we don't have to go
through this elaborate kind of
inference process there's a sense in
which like we just directly read off the
regularities in our sensory array uh
what what what is possible and what is
doable and it so in
gibson's original formulation this is
what
affordances meant
what we're doing is more drawing on tony
camaro's
reinterpretation of this concept as
affordances 2.0 he calls it where an
affordance is a relation
between some relevant features of the
environment and skills ultimately
you know expectations and skills had by
an organism so this is already a a more
inactive embodied reading of affordances
and i mean we've gotten into some very
interesting disagreements with people
from ecological psychology precisely
about this difference in interpretation
so i say this because umwelt is very
much focused on this kind of affordances
2.0 perspective where what we're trying
to do is really capture what the world

is like for the organism that inhabits
it from its perspective
this is also where i think it connects
with phenomenology which is also this
kind of more systematic attempt to
describe things from the first person
perspective

great

and

that okay alex first

and then shannon

yeah thanks um

as usual i'm trying to

connect uh to my domain is the

engineering domain

and

it's more like a question

about

could we consider importances from some

kind of functionalities view

as for example

if i need to drive in a nail

the hammer

the provide provides me affordance

to use it

but if i don't have a hammer

i could use even microscope to do the

same

only one or two times but anyway for my

engineering task it could be enough

and if i see a chair

and it's provide the affordance to sit

it's like a basic function of the chair

to

to to sit on it

so from

this perspective how it could be

considered like when agent

need to do something and need some kind
of functionality and this functionality
or maybe
service
to do it how it's possible link
oh absolutely this has been pursued by a
lot of work there's a really cool group
based in um
in the netherlands
uh eric rietveld and julia keverstein in
particular uh work from this perspective
where eric uh eric ridwell's work is in
architecture and affordances
so he he really tries to bring this
whole affordance's perspective uh into
the design of the spaces that we live in
uh so what you're saying is totally uh
consonant with uh at least some really
interesting uses of the concept of
affordances in the literature since the
early 2000s
cool shannon
i think yeah so
i'm just thinking of this this
field landscape difference and umvelt
and
it's really helpful considering the
perspective of the agent you're
investigating
especially um so i'm in music
neuroscience so we can learn a lot about
how
people or brains interact with music by
putting a person in an fmri scanner and
playing some music but that's not
exactly how we interact with music in
our day-to-day life maybe now more so
because we can turn on our ipod and just

sorry iphone we don't have ipods anymore
um and listen to music alone as this
um solo endeavor but in general for the
history of human evolution music has
been this really interactive
participatory
activity and in order to understand
um from where i am how we interact with
understand or process music it really
is also a question of how we understand
and interact with
um other people
and it's so when we bring in these
um
affordances we can have an affordance
for music um in a certain field that's
you know us listening to music alone but
as soon as the landscape
is
is a little bit bigger and maybe our our
small field incorporates one other
person or one more other person
um then the perspective that we need to
take
needs to change to adapt the perspective
of each of those individuals or of that
musical sort of environment as a whole
that's such a great point i don't know
if you're familiar with this paper that
i just linked
in the chat it's by anna anderson and
tom zimbardo
uh called exploring the multi-layered
affordances of context of composing and
performing interactive music with
responsive technologies
and it was one of the first papers to
pick up our cultural affordances work

and in precisely the context that you're
proposing shannon like um
what they're saying is uh our concepts
need to be enriched with a kind of
intermediate scale concept
so their idea is like when you're when
you're performing live electronic music
there's something intermediary between
the field of affordances that each
individual is coping with at any moment
and this evolutionary level kind of
landscape of affordances or this
population level uh landscape of
affordances where there's
you know the there are special
affordances that open up in this uh
performance
uh context
kind of a
a meso scale
in between the landscape and the field
uh i forget the precise term that they
that they introduced but uh yeah the
music is definitely one of the cooler
domains of application of this uh this
stuff
nice
sorry all right
shannon and then
just saying this is really great thank
you
cool sasha um yeah just a brief comment
to follow up on what alex said um
uh from the engineering perspective you
know if you have a hammer everything's a
nail it kind of made me think uh about
crime
where um if you need something and

you're not um
inhibited by uh law or social norms you
can just take it
and
maybe that's a uh
you know
overly simplistic understanding of
why people commit crime but in in that
context
if you're not inhibited by
uh
laws then your affordances are quite
different than if you are
okay and perfect richard
you're uh
yeah in terms of the the field versus
the landscape
would it be accurate to kind of
conceptualize this the field of force is
kind of the episodic knowledge of the
individual so what are their their
personal
learned experiences their personal body
capabilities kind of this really
episodic self you know um
form of memory where landscape of
performance is more about the semantic
knowledge the if there was a way for me
to kind of look at the population and
ask what are all the skills and habits
that this population has that would be
the landscape if i was to drill down on
one person
how well do you know how to ride a bike
that would be more of your fuel
performance precisely that's precisely
the idea and the idea also is that not
all affordances are equally salient at a

given time but they're still there in a sense right uh
so i don't know like um
24 7 uh coffee shop there's times a day where it's a salient affordance to some people well i was gonna precisely give a food and coffee related example uh yeah i i drink coffee in the morning but i tend to uh have insomnia if i if i drink it too late in the evening so there are times in the day when coffee affords drinking more than other times in the day and so you know it it's still the case that at any moment i could go into the kitchen and make myself a coffee right

so the field of affordances has more to do with those those affordances that are soliciting me at a given moment uh you know it corresponds to the more the phenomenology of interacting with affordances whereas the landscape is more an anthropological or sociological statement as as richard you were just saying about the kinds of things that a population can do in a given space

it
that you talk about that idea of sailings in the moment because one thing we talk about a lot in sociology is identity and so if you activate my identity prime me to think of as a professor i'm going to notice certain things in environment differently than if you prime me to think of my identity as a husband or a son or a brother um so that kind of then again but

identity means you know my identity as a professor may be different from any one of your identities as a professor based on socialization and culture and disciplines even as well right so okay interesting thank you yep and just one point to tie it back they write as high digger famously argued it is only when my smooth coping breaks down say when i run out of coffee or when the cut breaks that the objective properties the cup becomes salient present in perceptual experience at all and depending on if you're at home you run out of coffee maybe you go back to the coffee machine you set up a new brew if you're in a restaurant culturally maybe you raise your hand or maybe you whistle or maybe there's some other thing that's done culturally to signal that somebody else should see that as their affordance to come over and ask you what you need so the landscape of affordances is a very role specific uh it's realized through specific roles but it represents that population level and that's so yeah lots to say here that was an excellent discussion let's turn to these first two figures that are cartoons about basic cognitive formulas and a little bit more of a full formula so in this first three panel cartoon i'll read out the captions and then maybe somebody can give an example or a thought the first caption is the person thinking i think they think i think the second panel says what would others

expect me to do here and the third panel
says i inferred that they intend that i
should think feel or do x
so what is a thought or an
interpretation or a punch line to this
cartoon
what would mommy want me to do
in this context and i you know
it's i think this is
sociality is kind of scaling that up to
an anonymous kind of third
or other or
that is you know
yeah this kind of anonymous sociality is
kind of
scaling that
like kind of
you know parental expectation thing
up a bit
yeah it's like peer pressure made me do
it well yeah they move your joints for
you well no but people expected me to do
it well just because they expected you
to do it even if that's true why did you
do it and it comes back to this that is
an account
of what peer pressure is from from this
perspective even if the actual
biophysics of the joints are not being
pushed by other people and the the point
that we're making here i mean it's
really developed in our bbs paper uh
thinking through other minds but that
it's that human thinking really is not
just thinking about other humans but
thinking through their own perspective
uh and that much of our social
interactions actually are structured by

the beliefs that we have
about other people's beliefs about our
own behavior and beliefs
so you get this kind of nicely
recursive intentionality structure
uh
yeah
so that that's like the simple kind of
you know the basic cognitive formula
that we're calling it but the i we think
that this really applies to almost all
human thinking
we're really always kind of trying to
understand the situation
uh from the perspective of you know
shared beliefs about what's appropriate
in that situation which which implies
the ability to kind of perspective take
and to think through the minds of others
yep when someone's talking to hear
themselves talk as we say it's off the
rails because it's not in feedback with
other minds but when somebody is attuned
to the narrative and the culture of the
local world the local actual interaction
not just the hypothetical then there's
the opportunity for communication to be
targeted and direct and actually move
towards shared goals and under
shared values as well instead of just
individuals biophysically running their
mouth which is an affordance that we
also have so how do we move our
communication patterns towards this
higher level
um instead of just using the
opportunities that we know that we can
also have

any other thoughts on figure one figure
two is related
is that we talk a lot about
perspectivism and you know we go up to
higher and higher levels but i think
you see a point at which you can only
take a perspective on what you think to
a certain level it gets too complex too
multi-sensory too many inputs
and it is that kind of phenomenological
embodied feel for what you need to do
which is not something you even have a
perspective on that idea of the flow
so
there's this info inferential kind of
approach as well as a perspective
approach which just comes from
just feeling what you should do and i
think this this kind of shows that
because i think you can only go up so
far
and take perspectives on things
when
how you act in the moment sometimes you
just have to be there and feel it
yeah and we can't perfectly emulate what
somebody else wants and it's just not
that simple we're in the bodies that we
have with the affordances that we see
and so to try to over engineer what the
other person wants or should want or
should want to want it's uh it's a good
thought experiment and it helps in a lot
of situations but it is not the end game
of relating
well it might also explain some of the
difficulties that we're having currently
in our political landscape

uh so i mean as a as a canadian i'm very
sold on the idea of multiculturalism
and diversity and pluralism but i think
these are not intuitive
modes of sociality precisely because
uh they oblige us to multiply the
different perspectives
through which we're trying to
you know relate to a shared social world
um and you know like uh
things like dunbar's number i'm sure
you're all familiar with dunbar's number
right if you if you uh basically plot
the size of primate brains against the
size of their social groups there's
almost a linear relation and if you
extrapolate that to humans we would you
know live in groups of about 100
150ish
and you know the
the
that's that probably has a lot to do
with you know the the kind of sweet spot
where it's easy to kind of share
uh you know a bunch of uh beliefs and
norms and patterns of behavior with the
group and things get more and more
complicated as the group expands and
diversity is included which is not to
say that diversity is a bad thing by all
means i think uh you know we should move
in that direction
uh
but i think it speaks to some
difficulties that we're seeing
politically that it's not an intuitive
mode of sociality
cool

we'll return to that when we can
and here in figure 2 we're almost
rephrasing figure 1 a little bit more
from an improvisational perspective
so the left panel
the person is thinking how much can i
improvise here the middle panel
is what do i know they know about me
right now
and how much room to improvise does that
give me
the third
panel reads how much can i improvise
here given what the local cues tell me
about what others expect
so a lot to
say and think there's probably other
ways that these recursive questions
could be formulated but is there
anything that we haven't addressed in
one that people might want to address
here
cool this just reminded me of fellow
jitzer yes
yeah um
i was i was thinking that um
maybe in the adolescent period um
we are kind of rebel we are rebels
against society and trying to find
our identity
um
maybe in other words um minimizing our
self
prediction errors so we are like
trying to test
these norms um
by being uh level so actually by
impairing uh what others intend that i

should think and feel
maybe if i do the contrary
i'm going to
learn something about me
so
how can we deal with this um
like landscape affordances and cultural
affordances or even
what our mom
is saying and what we are really
um
trying to find out
um
about ourselves
nice maxwell
uh oh
well
uh yeah uh uh so i thought that was very
insightful uh thank you very much uh for
that very uh very interesting um
well i would have just added that so the
point here is to kind of say well
the
it's it's not it's not merely just about
you know inferring
uh what others think i should do it's
it's really more about the situational
coping right
so it's about like learning different
contexts
and learning about what's allowable in
all of these different contexts and then
being able to juggle that and to
reactivate the right constellation of
you know semantic knowledge for example
or rules and norms that pertain to each
context
and and also then you know moving from

one context to the next and seeing what
kind of flexibility that allows for
ultimately uh because you know human
reality is improvisational like it's
it's it's about kind of you know uh
coping with
uh
with a situation given the constraints
of that situation in real time
um yeah
yeah it's a controlled novelty question
like a lot of improvisation it's a on
the edge the leading edge of the
trade-off between explore and exploit
where successful systems are at and so
the teenage rebellion
if the teenagers just
mindlessly
fell into the line of whatever was
happening before
maybe in some niche that was unchanging
that would be a successful evolutionary
strategy but the niches are always
changing
not for the least of which is ecology
but also improvisation has this always
moving dynamic and so the teenage
rebelliousness is there's the element of
the individual coming to precision about
who am i and who am i in this broader
sense and then at the higher level
there's this explore exploit where some
element of contrarianism it comes across
like a negative word but if you've ever
had contrarian models then you couldn't
know if you were just in a very very
local optima or whether there are other
realms to explore

so there's there's so
much there and the details are all in
actually formulating these specific
situations and understanding what are
the affordances what is precision uh
doing here so maybe in the next bit we
can pretty
briefly look through figures three four
and five so
maxwell it will be helpful to have your
perspective on these but let's start
with three
so figure three is a hierarchical
prediction error minimization framework
and maxwell maybe could you give a shot
on what was being summarized here sure
um i mean we're just describing uh
bayesian predictive coding here
so
most uh
predictive coding schemes are
hierarchical as we were just discussing
uh above
where um basically what i mean this is
more explained in i think figure four or
five
but essentially what this figure in
particular is supposed to illustrate is
uh the kind of bi-directional message
passing that's implicit
uh in in all of this so
as we were discussing you know the top
or the center
uh is where the more kind of abstract or
slow regularities uh are represented and
they they effectively the the top layers
provide context for the bottom layers in
the same way that slow regularities

provide the context
or the embedding against which
uh you know faster regularities unfold
well these slower regularities or priors
provide context for the the ones below
so what you have descending or you know
coming from the center to the periphery
is effectively predictions about the
kind of
data that each
subsequent lower layer should be
receiving
and the way these schemes work
uh is uh i i think consonant with the
phenomenology
uh you know that so
essentially if the brain doesn't have to
process something then it doesn't uh the
brain is a lazy organ so the the thing i
i often say to drive this home is what
was the color of the last door handle
you opened
there's a large
there's a high probability that you
didn't register that at all
uh well because there's no reason to
right like you just smoothly coped with
with the door what you needed to do is
open the door to get to the other side
of it and you know there was no reason
to even register the color of the door
handle
so what with this kind of framework says
is the signal that the brain is
processing at any time is the
discrepancy between the data that it was
expecting to register and the data that
it actually does and this difference is

called a prediction error
and uh if you can go just back to figure
3 for one second um i just want to make
the point that well
so uh this relates to base in the sense
that descending messages
carry prior probabilities so essentially
you can think of this as the base rate
for a given phenomenon independent of
the data that i'm collecting and then
the prediction error kind of combines
that knowledge with data so it's
something like the likelihood if the
prior is just the probability of an
event that i'm considering the
likelihood is the probability of that
event given some data that i've actually
uh
registered effectively
so
this is why these frameworks are often
called bayesian the ascending messages
carry prediction errors which are
likelihoods and the descending messages
carry prior probabilities which which
are the predictions that the predictive
brain and so on are all about
perfect and then from here uh we move on
to
um a
specific hypothesis
the predictive coding i'm not sure 100
the degree to which we're still
committed to this functional distinction
but the idea is that um in at least in
the brain they're going to be two kinds
of neural populations
uh the ones which encode effectively

yeah these generative model units which are the ones that are encoding the the kind of base rates about the phenomena that you're interested in and then another uh functionally distinct sub-population of neurons that's responsible for computing the difference between the data you expected given these predictions and the data that you're actually receiving and again the unexplained signal that the part of the signal that wasn't predicted by uh the descending uh messages is passed upstream uh in the hopes that it'll be explained away at some point and also one note that can help understand this is this caption in the empirical base framework the system can then use the posterior obtained from one iteration as the prior in the next iteration and so the empirical base framework is in a way what it cuts the gordian knot it prevents us from just saying well it's just simply priors all the way up and we start somewhere by starting our prior for one level with just empirically what we're getting from the posterior at the lower level at that time point as a starting point and so this can be instantiated in physical not infinite systems because there's a possibility that empirical phase just as it's used in data analysis in a variety of fields you

can use the data to inform a prior and then have a generative model that starts working from there yeah precisely and what bayes does in this context is just provide a way to optimally combine what you knew before sampling any data uh with what you've learned from that data

uh so as you're saying uh what you're essentially doing is multiplying your posterior so you calculate your posterior by multiplying your prior by your likelihood and then uh normalizing uh which can get a bit hairy at times but

that's effectively what you're doing here

and let's just look at this example so here we have the same architecture as figure four so we're building it out figure three

this is a uh it's a graph it's just nodes and edges it's the pure representation now we're thinking about a little bit more of a plausible neural architecture and maybe in some brain regions it is this way and maybe in other brain regions it isn't this way but this is a plausible architecture and now we have an example so what is being shown with the dog

well this is just to re-illustrate the center versus periphery distinction uh well so the the image of the dog has just undergone a fourier decomposition which basically means that you're kind of separating out the different spatial frequencies

uh that are used to generate the image
and i mean the hypothesis is always sort
of the same is that the closer you are
to the sensory periphery the the more
you're dealing with faster changing
regularities uh so high spatial
frequency information is represented at
the bottom or the periphery of the
hierarchy and as you ascend the
hierarchy what you see is essentially
lower and lower uh spatial uh frequency
information

uh so this is the kind of idea here is
that like the the the fast regularities
are represented closer to the sensory
periphery slow ones closer to uh the
center

uh

yeah

i hope i said that right and um
and and yeah uh most sensory modalities
are gonna be

uh deconstructed in this way i mean you
know for the visual uh
brain regions

uh you start off you know detecting uh
essentially lines and you know like
grating patterns and the more primary
visual areas like v1 and everything
and then as you ascend the cortical
hierarchy what the neurons become
sensitive to are

groupings of these patterns so things
that change at a slightly slower
time scale

i mean if you did one of these fourier
decompositions to an image of the face
what you'll see is that the gross

morphological features of the face like
you know where your cheekbones are and
where your uh you know jawline is
are
much coarser than the fast changing
things in your face like your eyes and
your mouth
uh and so again you you might assume
that this is represented hierarchically
uh in the brain um
and let me see
i'll add one more level to the specific
visual example because it's in the
specifics that we can see what we're
getting out of this framework so low
spatial frequency is like a blur filter
it's a very uh it's averaged out over
large regions of visual space on in
contrast the very high spatial frequency
it looks grainy and that's kind of like
turning the contrast very high up to
over emphasize differences between
neighboring pixels but when you do that
you lose the global structure so we know
that from playing around in photoshop
well it would be really cool if we could
just see the world in total high
resolution and just optimize this with
one level but it turns out that in the
retina the cells are activated even
before the lines detection it's like
light first dark it can only detect
whether it's on or off extremely locally
and so how does that signal which is
ultimately an extremely local contrast
signal how does that percolate up in the
context of deep priors about
regularities in nature to look at deep

mind look at the images that are produced in deep dream these regularities that appear natural they're not just any kind of natural they're human visual niche natural like grass and clouds

these kinds of things have structures that certain other kinds of repeating uh like a grid just doesn't appear in nature and so this is a structure by which the hyper local contrasts to contrasts literally between pixels in this image can be passed up and you can end up getting this full spectrum for your uh resolution on the image and then to add one more level of detail what we experience isn't any single one of these levels because we experience a world in full color though there's no uh color receptors at our periphery and also a world that's at full resolution though we have very very different resolution as far as the pixels are uh the retinal detection units are concerned in this phobia in the center of the eye versus the periphery so that to me is demonstrative that we're experiencing our degenerative model of the world which is that the world has similar resolution no matter which way my head is looking and there's color everywhere and then subconsciously or sub personally our eyes are engaged in an oculomotor informational foraging task and where do the eyes secade it turns out that they look where the most informative place to

look would be
given the structure of this bayesian
algorithm so so many things that are
suggested with just this architecture
and there's so many aspects of the
signal processing cascade and especially
the top down priors that aren't
reflected by this idea that the eye is a
camera and the brain is a cpu and it's
just doing image classification
like a convolutional neural network
there are convolutional elements to be
sure in vision but they're playing out
in this type of framework not in a
tic-tac-toe framework
yeah absolutely
um
couldn't agree more with that that's uh
that's a great point i mean uh the i
think an often overlooked element of
this picture is that it's not just
saying
like this is a criticism of bottom-up
passive
you know a passive picture of the brain
that just kind of bottom up
aggregates uh you know simple properties
until you get something like a percept
it's so we're not just saying that this
is a top-down approach it's the
bi-directional processing cascade that
really gets you uh where you want to go
ultimately
and you know as you were just saying uh
dan like you know the the experience
that we're having
implicates all these different levels
and it's precisely because this whole

multi-level architecture is implicated
at all times that you get this kind of
rich
uh you know equally multi-level
experience
cool maybe we can turn to this last
figure for the last section of our
discussion so now we've built out this
bayesian framework from the previous
images first from its most austere
framing to something that looks a little
bit more naturalistic and now we're
going to bring it back to our discussion
about culture and about expectations
norms and affordances so here we have
the same
structure where we have predictions
moving
down and prediction errors moving up or
in or out these are just spatial
metaphors and there's only in and out
there isn't top or bottom
and then we're adding a few more levels
of details to the model the first is
this blue
and that's the precision weighting and
so that's entering this hierarchy at a
few different places and it's tuning the
temperature so to speak on the precision
so how much should we
value how much stock should we take
about this uh information that's coming
up although there's a lot more nuance
there and then also there's a top-down
modulation of attention and that's where
this regime of attention enters so the
precision story is really key to all
this and you know honestly as i was

learning this framework uh you know like five years ago initially i was hoping that the precision stuff wouldn't be so central to the argument because it's really complicated

but uh it turns out that if you get you get some of the most interesting effects from uh the precision stuff so it's it's it's a little bit

it's a little bit difficult sometimes to wrap your head around but i think the effort is worth it the the idea is just to say at every level of the hierarchy the the brain doesn't just register a signal but it evaluates how confident or reliable it is in that signal or how

reliable the signal is and this is what this precision construct effectively uh quantifies and you know uh dan you were relating this to temperature informally that's correct uh a more intuitive way perhaps of framing it is as volume so it's sort of like the the brain's volume on each of its different kind of sensory streams so sensory and predictive streams effectively

and yeah so uh precision weighting is what uh allows you to according to this story at least arbitrate uh different sources of information so for example 2018 2019

i spent a lot of time in london and london was surprisingly foggy

um so it it so far i mean as a canadian i never experienced anything like this i think once when i was a child there was just so much fog and there were days when i would come out of my flat in london and i couldn't see two feet ahead

of me there was so much fog and so uh
you know in in circumstances like that
what you learn to do is to rely on your
other senses like you're hearing for
example so there's a there's a story
about the way that the brain is kind of
dynamically it's sort of like a dynamic
balancing act
in a sense the brain is trying to see
which which of the signals that it's
dealing with are the most reliable um so
i'll stop now richard has yeah
yeah so and this is the the criticism i
get a lot uh when i present at
conferences and even when i publish on
ideas related to this in the sense of
you know if i'm let's imagine i'm a
master burglar and i i walk into a
neighborhood and i'm looking i'm chasing
the neighborhood to find out what i want
to what i want to break into which house
i want to break into and so i have these
priors i had the sensory information
highly motivated to steal things and
then there's always that old guy in the
back who raises his hand and says that's
really cool so how do i test any of that
how what do i measure what variables do
i need to collect and if at the end of
the day all you're giving me is a more
refined version of what i'm already
talking about
who cares and i i my gut tells me that's
wrong that i mean as we clean in all
these different disciplines we can go
back and forth and translate you know
our concepts back and forth and see that
we're all talking about the same thing

but if it doesn't lead to something
empirical that then we can then start
doing critical tests and show
this version of the theory is wrong
because my predictive priors or whatever
would have looked differently if it had
been that way
so i i guess like to me that's the thing
that i i'm always struggling with it and
i'm hoping i can get some insight from
everyone here is
how do we translate this figure six into
an empirical test that i can write up
for an nsf grant and then gather data
and show my theory is right
parsons from 1950 was wrong right or
something
that's a great question um
i think the where this has been done the
most extensively is in psychiatry
uh
where
and this is why you know uh this is what
really drove the message home for me
that the precision story was so central
to this
is that you can understand a lot of uh
psychiatric conditions uh in terms of uh
you know altered precision weighting
profiles
um
so
like uh i think a
kind of well understood
example has to do with the cycle the
course of schizophrenia
uh so we think that
and phil corlett uh has great work on

this uh i mean several people uh have worked on this uh from different kind of perspectives but all kind of relying on this construct of precision and backing it that then backing it up with fmri data so the course of schizophrenia for example it's thought to maybe start out as a you know a break a physiological break in your dopamine signaling so say d2 receptors start to misfire for no reason so they're generating uh what we would consider to be a spurious prediction error so like a prediction error that isn't really tracking anything and initially this gets resolved perceptually in the lower levels of the hierarchy it's kind of contained through hallucination like the brain is generating a prediction error signal and what it what it ends up doing is underweighting uh its priors are overweighting its prediction error uh because that's essentially what's going on like there's a breakdown and these prediction errors that aren't tracking anything are given too much uh importance and as the condition worsens they're given more and more importance because they're consistent like the brain is just consistently generating this prediction error and at some point it gets so it intense in the brain that it starts to be resolved through learning rather than through a hallucination so then you know for those of you who do a little bit of

research in psychiatry you'll know that there is a typical course of schizophrenia that starts off as a more kind of hallucination experience where people are still to some degree cognizant of the fact that they're they're undergoing an abnormal experience and they're hallucinating but late stage you know uh schizophrenia has more to do with uh persistent delusional beliefs and now your actual cognitive structure is adapting to the error signals that you're generating and learning from learning the error essentially uh which is not tracking anything in reality because it's just due to a breakdown so

from that point of view uh to answer the well to to address the points that you just raised richard the i think the uh like a lot of the power of the approach in terms of proof of principle comes from psychiatry these predictions uh and precisions are are computed by parts of the brain that you can identify and so phil corlett i think in particular has done uh some really really cool work uh on um on hallucinate uh on the uh on schizophrenia and the uh effectively the kind of uh presentation of schizophrenia and the chorus of schizophrenia very cool work from that point of view it feels at yale uh for those of you who are interested cool um steven and then we'll return to

the unique

predictions question

yeah i suppose just adding to that um

this this challenge of

things happening over time like a time

series

of

things happening rather than kind of

some snapshot in time which i think

traditionally a lot of research sort of

does that

so

even what maxwell was saying there is

what's presented as a pattern over time

if you took one moment it might not show

that and i think that's where

it's kind of hard to analyze that that's

why these models that they create

show how things pan out over time

on a low dimensionality and then you can

get something from that and i'm kind of

that does present a problem about real

life scenarios and i'm my i'm interested

in whether maybe

approaching it from the direction of

affordances and the environment and

working backwards

to how people are tuned to an

environment might be the way to look at

the more macro because it becomes so

high dimensional if you take someone at

the individual level but i think this is

up for grabs

let's bring that back to the question

about crime uh by richard so let's think

about the shared expectations so the

shared expectations might be people

don't walk uh onto someone else's

private property so in that case even somebody who's highly motivated and has the tools the affordance to do a burglary there's still uncertainty about what's inside the house now let's imagine changing the norm so that people are have wide windows and you can see exactly which house has which uh equipment now somebody who's motivated and with the affordance is enabled by a cultural norm that allows passing close to the window as well as an architectural affordance to see inside the house they're reducing their uncertainty about what's inside of the house and that is going to change not just that agent's behavior in their own field of affordances but also some macro trends and so this is in one sense this is looking at within the brain these are different neuronal populations and that would be precision could be played by uh dopamine and by dopaminergic synapses like maxwell's talking about and personally i find it fascinating that so many stories about reward like oh well gambling is about people trying to maximize reward or bank robberies about people trying to maximize reward really is it though or is it about people optimizing their precision or perhaps even operating under maladaptive uh precision regimes so it doesn't have just because it's an optimization and a precision oriented framework doesn't mean that aberrant events don't happen

so within the brain level we can think about that type of precision but then also socially what is the feedback that somebody receives along the way that have has other people give them feedback to keep them uh

on the straight and narrow as it were and then how do we as cultures define what that path is are there on-ramps from the fringes of society that help onboard people into a healthy and cooperative way of living or do we push people who even deviate in some way out of the center where they're even more likely to deviate further from the increasingly narrow perspective of the center so there's really so many levels that this can come into play at so i think just because of the time

let's have a few closing thoughts from anybody who would like to share and then also please through text if you're listening as well as participating submit us questions or topics or quotes because we have a bunch of quotes that we've selected and some other things that we felt like we could have spent more time talking about but we'll save that for next week with this first week's podcast being the figures the definitions the on-ramps and the accessibility from a lot of different fields and then the next week we'll bridge out even further so any closing thoughts from our participants

shannon

i think that's a really good point that

question of how do you
study this you know how how is how do
you
bring this entire story of
the free energy principle or predictive
processing or act of inference and
actually study it in
a
social context like in a group of people
so not just the stationary burglar who's
alone in a neighborhood
at night but in the context of
a group of people who are interacting
and i think that would be a really great
topic for part of next week or like when
we revisit
and just you know a plug from music
that's a great uh
little model system to study these kind
of interactions um and starting from
western music just because i grew up
here and that's kind of music i've
interacted with but if you have an
ensemble you know who are the leaders of
the ensemble who are the followers
and you can kind of manipulate that um
and
really study how you can play with
changing the regimes of attention or
changing who in the group
dictates uh what the particular field
of affordance is that this piece of
music needs to play
um and there are a lot of really
exciting and fun ways to get creative in
the lab to study these kind of social
exchanges and maybe put some of these
um super complex mathematical models to

like bring it give it some empirical testing

great richard anyone else can raise their hand as well yeah uh building off that i think that's a great idea to talk about people in groups and you know the things that i've seen coming out with with hyperscanning and the ability to have two three four or five people all being scanned at the same time whether socially interacting and and then asking okay so what is this mathematical model of the free energy principle or active inference or whatever embedded an individual how does it what i mean does it come down to a couple of oscillators i mean that's the thing i keep coming back to you is this right this is just another version of coupled oscillators and if that's true

then the notion of how do i empirically deal with this is because these coupled models can deal with 10 15 20 1 000 people interacting that our current concepts in sociology we can talk about abstractly but no one can really figure out a way to measure it so i think that'd be really interesting to look at the intersection between the models

our our current scanning abilities and empirical design and the theory and really see where they what are they talking about and what is a more unified you know

operationalization of these concepts in light of this technology and these

models what would that look like in
terms of experimental design
that's really exciting um i'll i'll try
to rejoin uh because my camera crashed
again after this comment but i wanted to
say to both of you shannon and richard
if these are lines of research that
you're interested in exploring this is
definitely something that my group is
very keen on exploring uh i mean you're
talking about hyper scanning i really
think this is the direction ford uh
we're working with uh
a few people now uh
so robin murphy
uh santiago castillo uh
in particular
are all
people who do eeg and fmri hyper
scanning and uh the the new kind of
thing that we're trying to do is uh to
combine multi-agent modeling with uh
multi-brain neuro imaging
you know because the problem with that i
have with social neuroscience is that
we're and i'm sure those of you who do
social sciences will concur
they're what we're essentially doing is
looking at the brain with this really
deep level
of you know inquiry we're
you know
we're looking at like hundreds of
different uh sensors on the scalp or
whatever and so we have all of these
this brain data that we're looking at
but really when we're looking at like
the kind of social measures that we're

using it's it's these kind of toy
measures like empathy or whatever that
are supposed to tell us like something
broad about culture
and the problem with that is that we're
effectively flattening all of the causal
structure and complexity of the social
world onto these kind of toy dummy
variables that are supposed to tell us
something really deep about uh you know
social interactions and
uh by combining this multi-scale
perspective opened up by the free energy
principle with uh hyper scanning and
other uh social neuroscience techniques
we're hoping to really
be able to uh talk about and understand
and model the social world in all its
complexity while also looking at
uh brain dynamics so i have people for
example interact in ecologically
realistic situations for humans
uh and then see stuff about you know
like uh so edwileck for example if you
don't know her you should absolutely
check out her work um
e-d-d-a-b-i-l-e-k
edward
she's currently carl fristan's postdoc
uh and i think she's also at mannheim in
germany and what she's developed is uh
effectively a measure of brain to brain
synchrony or uh brain tumor real-time
brain-to-brain information transfer
and you can get to this by uh looking at
basically real-time interactions between
humans
uh you know engaging in some kind of

shared participatory task like you know
aligning gaze direction or whatever and
simultaneously recording their brains as
they interact and i really think that
this is like the golden way forward so
if any of you actually are interested in
pursuing this please uh send me a dm or
something and we can get started
perfect and i'll just close i think with
one short note on flattening versus
deflationary theory because it really
does come back to what are we doing here
is this another metaphor and it's just
pendulums on a table um or is there
something actually more here and
flattening maxwell was suggesting is
when we're just projecting that's
psychological as well as a dimensional
term we project onto simple constructs
we want to have a simple concept of
empathy and in a sense by projecting
into that representation at a lower
dimension we've created a world that
isn't necessarily reflective of the
whole system
in uh in contrast with a flattening
account we can have a deflation a
deflationary account where what's
happening is really communicating brains
it's really just those brains
communicating and we're deflating away
the extra the second level where is the
empathy where's the regime where's the
attention the answer is it's the whole
system and it's not just the whole
system meaning the person it's the whole
system with the niche and the culture
and so it's a lot to grasp grasp with

because it is a multi-level system and we can design an experiment that isolates a certain component but we never truly have isolated from the evolutionary context or the developmental context and so the goal to pursue inflationary rich accounts that provide all of these humanistic things that we want to see like inactive embodied in cultured cognition rather than flattening it onto something and then saying well maybe they're the same between different cultures we can actually

go into that richness and look at what we actually have and stephen with a closing thought

yeah thanks i totally agree and i like this idea of being able to feel where we look this idea of feeling texture it would be sort of analogous to that because you have empathy and someone like i work a lot in the development field and community work and you get this idea of empathy but there's no

time variation and empathy it's just like dump there resistance and it's like well how is that changing moments moment isn't there's none of that no one really thinks about that how things change over time so i think this this is actually really useful it's clarified a lot of stuff in my head actually just hearing that discussion about

i think it's a different way of thinking about

looking normally all that granularity
would be seen as a problem but actually
it can be seen as data which is really
cool

beautiful maxwell first author last
thought

i just wanna it's less of a thought than
a feeling i just wanted to re-express my
gratitude to everyone in the
conversation um this is really exciting
and

awesome and uh like yeah i i
so

if anyone does want to develop these
ideas like i was saying please get in
touch with me and uh our our group is
very keen on collaborating with the
we've established a a multi-site
international network of research called
the nested minds network uh we're gonna
give uh we're gonna launch the website
pretty soon but uh for those of you who
uh are looking for international uh
collaboration and so on please get in
touch with me uh

like what we're trying to do is build a
community uh around this and you know
this podcast is definitely part of the
consolidation process
and uh yeah so thanks thank you all uh
you're all really cool and this is
really fun

thank you maxwell we built the on-ramps
together that are bigger than we can
build individually and build the
on-ramps we want to see we build the
culture that we want to see and that's
a lot of work to be done but it's

empowering to know that together we'll
make it happen
okay thank you all for the live stream
thanks for listening and i will end the
live stream